

<https://www.rtl-sdr.com/rtl-sdr-quick-start-guide/>

- just follow

<https://www.rtl-sdr.com/sdrsharp-users-guide/>

After opening SDR# for the first time, we suggest that you immediately remember to perform the following steps (if you don't know what some of these steps are, continue reading further below for more information):

1. Increase the RF gain from zero to a higher value in the configure menu.
2. Reduce the range slider on the right of the SDR# window to about -70 (for RTL-SDR dongles).
3. Enable the "Correct IQ" setting to remove the center spike if using an R820T/R820T2, or enable "Offset Tuning" in the configure menu if using an E4000/FC0012/13.
4. Turn off the "Snap to grid" setting, or adjust the PPM offset accordingly.
5. Set the 'Mode' to the correct setting for the signal that you are listening to.

| Mode           | Acronym Expansion                | Explanation                                                                                                                                                                                          |
|----------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NFM</b>     | Narrowband Frequency Modulation  | Commonly used mode used by walkie talkie radios, weather radio and most VHF/UHF digital signals.                                                                                                     |
| <b>WFM</b>     | Wideband Frequency Modulation    | This is the mode that broadcast FM stations use (e.g. radio music stations).                                                                                                                         |
| <b>AM</b>      | Amplitude Modulation             | Used by broadcast AM stations that are receivable by normal shortwave radios and also used by air band voice frequencies used by aircraft and air traffic control. Some digital signals also use AM. |
| <b>LSB/USB</b> | Lower Side Band / Upper Sideband | Used in the HF band by ham radio users to transmit voice and data efficiently with small bandwidths.                                                                                                 |
| <b>CW</b>      | Continuous Wave                  | Used for listening to Morse code.                                                                                                                                                                    |
| <b>DSB</b>     | Double Side Band                 | Similar use to AM, but requires centered tuning.                                                                                                                                                     |
| <b>RAW</b>     | Raw IQ signal                    | Used for listening or recording RAW IQ data                                                                                                                                                          |

- must select for it to demodulate properly

| Setting                         | Suggested Default          | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Shift</b>                    | 0 (No upconverter used)    | This box offsets the tuned frequency by the amount entered. This is useful if you are using an upconverter. For example if you have an upconverter with a 100 MHz oscillator, you would set the shift to be -100,000,000 (don't forget the minus sign). Without the shift, when using an upconverter to tune to a signal at 9 MHz you would need to actually tune to $100 + 9 = 109$ MHz. With the shift set, you can tune to 9 MHz as normal. If you have an upconverter with a 125 MHz oscillator you would tune to $125 + 9 = 134$ MHz, or set the shift to -125,000,000. |
| <b>Bandwidth</b>                | NFM/AM: 12500, WFM: 250000 | This is the width of the shaded part of the tunable area. You can set it manually here, or by dragging the edges with the mouse as described under the tuning bar description.                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Filter</b>                   | Blackman-Harris 4          | Changes the filter type used. Different filters have different shapes. The filter is used to select the highlighted signal in the RF window. Blackman-Harris is usually the best filter to choose and this setting almost never needs to be changed.                                                                                                                                                                                                                                                                                                                         |
| <b>Order</b>                    | 500                        | Changes the filter order. You may notice when using low filter orders that signals outside of the tuned bandwidth can still be heard. Larger filter orders “tighten” or “sharpen” the band pass filter used within the tuned bandwidth/IF thus preventing signals outside of the tuned bandwidth from being heard. You will want to increase the filter order when there are strong signals near to your tuned area. Using higher filter orders can cause a greater load on the CPU, so slow PCs may need to reduce this value.                                              |
| <b>Squelch</b>                  | OFF                        | Squelch is used to mute the audio when the signal strength is below the specified value. A larger value requires a stronger signal to unmute the audio. It is useful for when listening to speech as the sound of static when no one is talking will be muted.                                                                                                                                                                                                                                                                                                               |
| <b>CW Shift</b>                 | 600                        | Mainly useful for when receiving CW (Morse code) as it specifies the offset between CW transmit and receive frequencies.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>FM Stereo</b>                | OFF                        | Will enable stereo output for broadcast radio WFM signals, but can make weak stations sound worse.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Snap to Grid / Step size</b> | OFF or set to band spacing | In many bands frequencies are allocated at a fixed distance apart. For instance in most countries air band signals are spaced 25 kHz apart (or 8.33 kHz in some countries). Turning on snap to grid can help with tuning by causing the tuning bar to snap directly to a signal. However, to use this with a non-TCXO RTL-SDR the PPM frequency offset correction must be set correctly, otherwise the frequencies may not line up. The snap to step size can be set in the “Step Size” pull down menu.                                                                      |
| <b>Correct IQ</b>               | ON                         | Should usually be selected as ON. This setting removes the small but annoying center spike that is present with R820T/R820T2 RTL-SDR dongles.                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                       |     |                                                                                                                                                                                                                                 |
|-----------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Anti-Fading</b>    | OFF | Can be used when “Lock Carrier” is activated. Takes advantage of the symmetry of AM signals which helps with weak signals when they may be fading in and out. Turn this on for better AM reception, but may increase CPU usage. |
| <b>Swap I &amp; Q</b> | OFF | If you are using SDR# as a panadapter, some hardware radios may have the I & Q signals swapped and need this option checked.                                                                                                    |
| <b>Mark Peaks</b>     | OFF | Simply marks any peak in the RF spectrum with a circle.                                                                                                                                                                         |

- standard setting